

Homework 12

Arham Lodha

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Problem 1a.

Proof. $\forall \epsilon > 0$. Let $\delta = \epsilon/2$. $\forall x, u \in [1, \infty) \ni |x - u| < \delta$.

$$\begin{aligned} \left| \frac{1}{x^2} - \frac{1}{u^2} \right| &= \left| \frac{u^2 - x^2}{x^2 u^2} \right| \\ &= \left| \frac{(u - x)(u + x)}{x^2 u^2} \right| \\ &= \delta \left| \frac{u + x}{x^2 u^2} \right| \\ &= \delta \left| \frac{1}{x^2 u} + \frac{1}{x u^2} \right| \\ &< 2\delta \text{ (since } x, u \geq 1) \\ &= \epsilon \end{aligned}$$

Thus $\frac{1}{x^2}$ is uniformly continuous on $[1, \infty)$

□

Problem .

Proof. Let $\epsilon_0 = 0.5$. Let $(x_n) = \frac{1}{\sqrt{n}}$ and let $(u_n) = \frac{1}{\sqrt{n+1}}$. x_n and u_n converge to 0. By operation of limits, $\lim_{n \rightarrow \infty} (x_n - u_n) = 0$. However $\forall n \in \mathbb{N}$,

$$\begin{aligned} \left| \frac{1}{x_n^2} - \frac{1}{u_n^2} \right| &= \left| \frac{1}{\frac{1}{(\sqrt{n})^2}} - \frac{1}{\frac{1}{(\sqrt{n+1})^2}} \right| \\ &= |n - n - 1| \\ &= 1 > \frac{1}{2} \end{aligned}$$

Thus by Nonuniform convergence criterion, $\frac{1}{x^2}$ has uniform convergence on $(0, \infty)$.

□

Problem 2.

Proof. By the definition of a closed set, $I = \bar{I} = I \cup I'$ and I' is the set of limit points of I . Thus I has the set of limit points of I . Thus since x_0 is the limit of a convergent sequence $(x_n) \subset E \subset I$, $x_0 \in I$.

By the definition of sequential definition of continuity, since $x_n \rightarrow x_0$, $f(x_n) \rightarrow f(x_0)$, and $g(x_n) \rightarrow g(x_0)$. Thus, $\lim_{n \rightarrow \infty} f(x_n) = f(x_0)$ and $\lim_{n \rightarrow \infty} g(x_n) = g(x_0)$, however since $\forall n \in \mathbb{N}$, $x_n \in E$ thus $f(x_n) = g(x_n)$. Thus $\lim_{n \rightarrow \infty} g(x_n) = f(x_0)$. By the uniqueness of the limit, $f(x_0) = g(x_0)$. Thus $x_0 \in E$. $\square$

Problem 3.

Proof. Let (x_n) be the following recursive sequence. $x_1 = a$. $\forall n \in \mathbb{N}$, $\exists y \in I \ni |f(y)| \leq \frac{1}{2} |f(x_n)|$, let $x_{n+1} = y$. Thus by construction, $\forall n \in \mathbb{N}$, $|f(x_{n+1})| \leq \frac{1}{2} |f(x_n)| \leq \dots \leq \frac{1}{2^{n-1}} |f(x_1)| = \frac{1}{2^{n-1}} |f(a)|$.

$(x_n) \subset I \implies \forall n \in \mathbb{N}, a \leq x_n \leq b$. Thus x_n is bounded. By Bolzano Weierstrass theorem, $\exists y_n$ a subsequence of x_n which is monotonic, since x_n is bounded. Since y_n is a subsequence of x_n it is bounded as well, by monotonic subsequence theorem, y_n is a convergent sequence. Thus $\exists c \in \mathbb{R} \ni y_n \rightarrow c$. By the definition of a closed set, $I = \bar{I} = I \cup I'$ and I' is the set of limit points of I . Thus I has the set of limit points of I . Thus since c is the limit of a convergent sequence $(y_n) \subset E \subset I$, $c \in I$.

Since $\forall n \in \mathbb{N}$, $y_n \in (x_n)$ and by construction, $\forall n \in \mathbb{N}$, $|f(y_{n+1})| \leq \frac{1}{2} |f(y_n)| \leq \dots \leq \frac{1}{2^{n-1}} |f(y_1)| \leq \frac{1}{2^{n-1}} |f(a)|$.

Thus, $0 \leq |f(y_n)| \leq \frac{1}{2^{n-1}} |f(a)|$.

$\forall \epsilon > 0$. By the Archimedian property, $\exists K \in \mathbb{N} \ni K > -\log_2(\epsilon)$. $\forall x \geq K$.

$$\begin{aligned} \left| \frac{1}{2^n} \right| &= \frac{1}{2^n} \\ &= \frac{1}{2^K} (\text{since } n \geq K \implies 2^n \geq 2^K) \\ &< \epsilon \end{aligned}$$

Thus $\lim_{n \rightarrow \infty} \frac{1}{2^n} = 0$. By operation of limits, $\lim_{n \rightarrow \infty} \frac{2}{2^n} |f(a)| = \lim_{n \rightarrow \infty} \frac{1}{2^{n-1}} |f(a)| = 0$. Limit of constant sequence is constant, thus 0 converges to 0. Thus by squeeze theorem, $\lim_{n \rightarrow \infty} f(y_n) = 0$. By the sequential definition of continuity, $\lim_{n \rightarrow \infty} f(y_n) = f(c)$. By the uniqueness of the limit, $f(c) = 0$. Thus $\exists c \in I \ni f(c) = 0$. $\square$

Problem 4.

Proof. Let $f(x) = x$, $g(x) = c$. $\forall r \in \mathbb{R}$. $\forall \epsilon > 0$, let $\delta = \epsilon$. $\forall x \in \mathbb{R} \ni |x - r| < \delta$. Thus $|f(x) - f(r)| = |x - r| < \delta = \epsilon$. Let $\gamma = 1$. $\forall x \in \mathbb{R} \ni |x - r| < \gamma$. $|g(x) - g(r)| = |c - c| = 0 < \epsilon$. Thus, $f(x), g(x)$ are continuous at $\forall r \in \mathbb{R}$. Thus $f(x), g(x)$ are continuous on $\mathbb{R}$. $\forall n \in \mathbb{N}$, $a_n x^n = a_n (\prod_{k=0}^n f)(x)$, thus by operation of limits, $a_n x^n$ is continuous. Thus $p(x) = \sum_{n=0}^{\deg p} a_n x^n$ is continuous by operation of limits.

Thus $p(x) = x^4 + 7x^3 - 9$ is continuous. $p(0) = -9$, $p(2) = 63$, $p(-10) = 10^4 - 7(10^3) - 9 = 10000 - 7000 - 9 = 3000 - 9 = 2997$. By the intermediate value theorem, $\exists a \in (0, 2) \ni p(a) = 0$ and $\exists b \in (-10, 0) \ni p(b) = 0$. Since $(-10, 0) \cap (0, 2) = \emptyset$, a, b are distinct. Thus p has at least 2 roots.

(Note an alternate proof would use FT of Algebra and the fact complex roots come with its conjugates, so it suffices to show there is at least 1 real root.) $\square$